



Assignments Overdue: You can still pass! Remember, you need to pass these assignments before the course ends on July 9, 11:59 PM PDT.



THIS WEEK'S FORUM

Week 4

Discuss this week's modules here.

[Go to forum](#)

Count data and hierarchical modeling



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Poisson regression, hierarchical modeling

Learning Objectives

Specify, fit, assess, and use common statistical models for count data.

Extend basic statistical models to account for correlated observations using hierarchical models.

Use statistical modeling results to draw scientific conclusions.

Module Overview



Module 4 assignments and materials 3 min

10. Poisson Regression



Introduction to Poisson regression 3 min



JAGS model (Poisson regression) 17 min



Predictive distributions 11 min



Quiz:

Lesson 10 8 questions Due June 25, 11:59 PM PDT



Prior sensitivity analysis 20 min



Code for Lesson 10

11. Hierarchical Modeling

-  Correlated data 8 min
-  Normal hierarchical model 20 min
-  Prior predictive simulation 10 min
-  JAGS model and model checking (hierarchical modeling) 13 min
-  Posterior predictive simulation 8 min

 **Quiz:**
Lesson 11 Part A 8 questions Due June 25, 11:59 PM PDT

-  Linear regression example 7 min
-  Linear regression example in JAGS 10 min

 **Quiz:**
Lesson 11 Part B 6 questions Due June 25, 11:59 PM PDT

-  Applications of hierarchical modeling 10 min

 **Discussion Prompt:**
Selecting prior distributions 5 min

-  Code and data for Lesson 11

Optional Honors Content

Mixture Models

-  Mixture model introduction, data, and code 20 min
-  Mixture model in JAGS 13 min

 **Quiz:**
Predictive distributions and mixture models 5 questions Due June 25, 11:59 PM PDT